

The Performance of a 2018 Mazda CX-5

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Abstract

In this epic quest, the performance of the 2018 Mazda CX-5 was analyzed using Veepeak OBD-II data at a sampling rate of 10 Hz. The vehicle performance data was collected across 3 different trails, capturing periods of idling, acceleration, cruising, and deceleration. The recorded variables include time, speed, acceleration, engine RPM, fuel rate, and distance traveled. The data was used to develop mathematical models describing position, velocity, acceleration, RPM, and fuel consumption as functions of time and speed across different courses. These courses included maintaining 15 mph for 100 yards, maintaining 25 mph for 200 yards, driving a quarter mile at an average speed of 15 mph, and a quarter-mile drag strip.

Introduction

The goal of this epic quest is to create mathematical models of four different courses and compare them with the experimental data collected using the OBD II reader. The vehicle's performance, including acceleration, velocity, position, fuel consumption, and engine RPM, was recorded. This data allows us to study and model the vehicle's dynamics virtually. In this project, we used a Mazda CX-5, a popular vehicle representative of commonly driven SUVs that operate in low-speed, stop-and-go conditions. Studying these conditions enables us to observe how acceleration, braking, and idling impact the car's engine behavior and fuel consumption.

We conducted multiple driving trials in a closed course to ensure repeatability and safety. During these 3 recorded courses, we collected data on the vehicle at 10 hz. These measurements were selected to allow the construction of mathematical models describing the position, velocity, acceleration, engine speed, and fuel usage as functions of time and speed. This project demonstrates how kinematic equations can be applied to real vehicle data while highlighting the uncertainties of the OBD reader.

Methods and Backgrounds

The goal of this project is to develop a mathematical performance model for a personal vehicle using experimentally collected driving data. The model is designed to predict acceleration, braking behavior, engine performance, fuel usage, and power output under real-world driving conditions. Experimental data were collected using an OBD-II reader and a smartphone accelerometer, allowing direct measurement of vehicle speed, engine RPM, acceleration, and fuel-related parameters.

To ensure the model currently in place is valid and can also be repeated, our data collection followed a controlled procedure using multiple runs of the same driving tasks. Each model output of 0 to the target speed time, braking time, RPM vs speed by gear, fuel consumption, and power vs speed, is based on at least three datasets and then processed in MATLAB using numerical differentiation/integration, and curve fitting. The final model is built so within the tested range it can be interpolated and extrapolated further to predict performance on longer unknown courses, including an Irwindale drag-strip simulation.

Step-by-step procedure for data collection workflow

1. Equipment

- 2018 Mazda CX-5
- Bluetooth OBD-II reader
- Recording device (iPhone)
- OBD Fusion (\$10)
- A nice area to drive safely and reach the required speed between 0 and 35mph.

2. Vehicle preparation

- Prepared the vehicle to normal operating conditions to reduce variation in RPM behavior and fuel rate
- Verified safe test location: straight roads with minimal traffic

3. OBD-II setup

1. Place the Veepeak device under the steering column of the car, beneath the driver's foot area.
2. Plug the OBD-II reader firmly into the port.
3. Start the engine.
4. Enable Bluetooth on the mobile device.
5. Go to settings.
6. Select the OBD-II device from the list of available devices.
7. Open the OBD Fusion app.
8. Go to settings and edit the necessary units.
9. Select PID's and toggle them for each trial.
10. Go to settings: vehicle profile: add profile
11. Enter the vehicle: Mazda 2018
12. Torque App should now display miles of distance traveled, mpg, speed, acceleration rates, etc.

4. Acceleration test: 0 → target speed mph

- Started logging on OBD and accelerometer.
- From rest, accelerated smoothly to 35 mph.
- Held near target speed mph briefly to capture a stable endpoint.

- Stopped logging and saved the run.
- Repeated until 3+ runs were collected

5. Documentation

- Recorded an external video for multiple data trial runs.
- Analyzed and logged run notes: driver and speed consistency, time of run, ensuring the target distance was reached

To model the vehicle's performance, we had to collect data on speed, acceleration, distance, engine RPM, and fuel rate, collected using the OBD II. Time was measured as the independent variable. Speed and acceleration were measured to determine and calculate kinematic equations. We measured multiple scenarios, including idling, cruising, accelerating, and decelerating. The engine rpm was collected to create a relation between engine behavior to speed. Fuel rate was measured to evaluate efficiency and total fuel consumption during different driving conditions. All the data was recorded at 10 Hz for consistent and accurate data.

During our data collection, the main challenge was finding an iPhone app that would reliably connect to the OBD II adapter while allowing access to the required variables and sampling rate. Several applications were insufficient, such as Torque Pro and Car Scanner, so OBD fusion was selected to meet the requirements. There was additional time needed to learn the interface and export the data in CSV format for analysis.

Vehicle Motion Modeling

Vehicle motion is described using basic kinematics. Position, velocity, and acceleration are related as:

$$\text{Position} = s(t)$$

$$\text{Velocity} = s'(t)$$

$$\text{Acceleration} = v'(t)$$

Acceleration data obtained from the smartphone accelerometer were numerically integrated in MATLAB to estimate velocity and position, while OBD-II speed data were used to validate the results. Multiple datasets were averaged to reduce messiness and better repeatability. Polynomial and exponential curve-fitting were applied to allow interpolation and extrapolation past the measured range. These models were selected to allow for reliable interpolation of the measured data points for performance prediction while avoiding noise or non-physical behavior.

Acceleration and Braking Models

Acceleration from rest to the target speed in mph was modeled using velocity vs. time data collected during the controlled trial test runs. Time to speed predictions, like the time of 0-20 mph, were obtained by solving the fitted velocity mathematical model for a given speed.

Braking performance was modeled using deceleration data measured during stops from 15 mph to rest. Assuming approximately constant braking force over short durations, braking time from any initial speed was estimated using the physics formula that $\text{time} = \text{initial velocity} / \text{acceleration}$ or $t = v_0 / a$

Engine RPM, Gearing, and Power Band

Engine RPM was directly measured using OBD-II data and correlated with vehicle speed to identify gear-dependent relationships. Each gear produces an almost linear relationship between RPM and speed due to fixed gear ratios.

The power band concept, the RPM range where the engine produces maximum usable power, was used to justify acceleration behavior and shifting. While torque was not directly measured, power cannot be calculated through physics, but power vs. speed trends were estimated using fuel rate, acceleration behavior, and RPM data trends.

Fuel Consumption Modeling

Fuel usage was estimated using OBD-II fuel rate data. Total fuel consumption for a given driving course was determined by integrating the fuel rate over time. This allowed the prediction of fuel usage for unknown courses and extrapolated driving scenarios.

Extrapolation to Track Conditions

Using fitted models, vehicle performance was extrapolated to simulate behavior on the Irwindale Speedway drag strip. Assumptions included consistent road conditions and steady engine performance. Acceleration models were extended beyond the 35 mph dataset using polynomial and exponential fits, confirmed with known physical behavior.

Technical Issues

- Repeated disconnect of the OBD-II data reader to the mobile device

- Fusion OBD App malfunctions in data collection
- Improper calibration of the OBD to a frequency of 10 hz

Results

Table 1. Summary of OBD II Data Used for Modeling

Variables	Units	Source	Purpose
Time	s	OBD data timestamp	Independent variable
Vehicle speed	mph	OBD speed PID	Velocity, RPM vs. speed models
Engine RPM	rpm	OBD RPM PID	RPM vs speed, RPM vs time. models
Fuel Rate	gal/hr	OBD fuel PID	Fuel rate model
Distance	miles	Derived from OBD speed, used OBD PID for reference	Course segmentation
Acceleration	ft/s ²	Derived from speed data, used OBD PID for reference	Model validation

Table 2. OBD II data used to construct vehicle models.

Couse	1	2	3
Time range	0:12.201	0:22.905	0:63.905
Vehicle speed range	0:15.534	0:21.748	0:18.641
Average speed	10.472	15.845	14.126
Engine RPM range	0:1792.2	0:2641.5	0:2385
Average Engine RPM	1293.6	1592.3	1196.7

Fuel rate range	0:1.394	0:2.4681	0:2.2654
Average fuel rate	0.87581	0.90235	0.57398

Uncertainty

Due to the use of consumer-grade OBD II sensors, all the measured variables had uncertainty by default. The vehicle speed data had small fluctuations due to the sensor's resolution and introduced noise into the data. One main source of uncertainty was a malfunction within the OBD reader during the conduct of each course trial. Lack of a secure connection caused continuous resets and reconnects of the OBD to our mobile device. This may be a source of uncertainty as the OBD data collection is not technically calibrated, so different trials may see variations in the precision or accuracy of the data collected. Also, since the track was not fully flat, it could have caused uncertainty. To reduce random error, each course was repeated three times and averaged prior to model fitting. While this approach does improve the repeatability, there is still some uncertainty that remains, and the performance should be interpreted as an approximation rather than exact values.

Figure 1. Position vs. Time

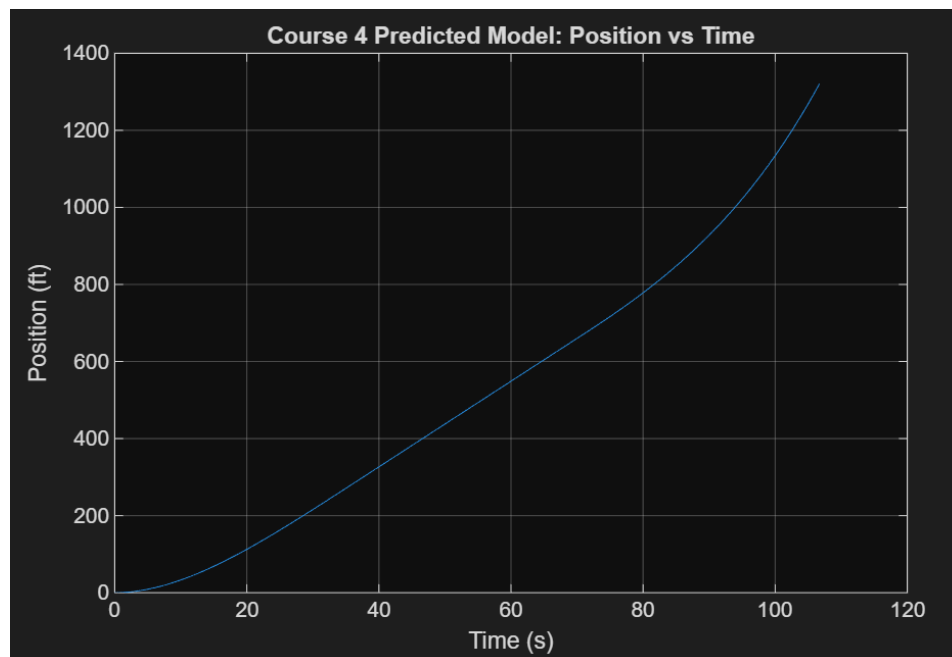
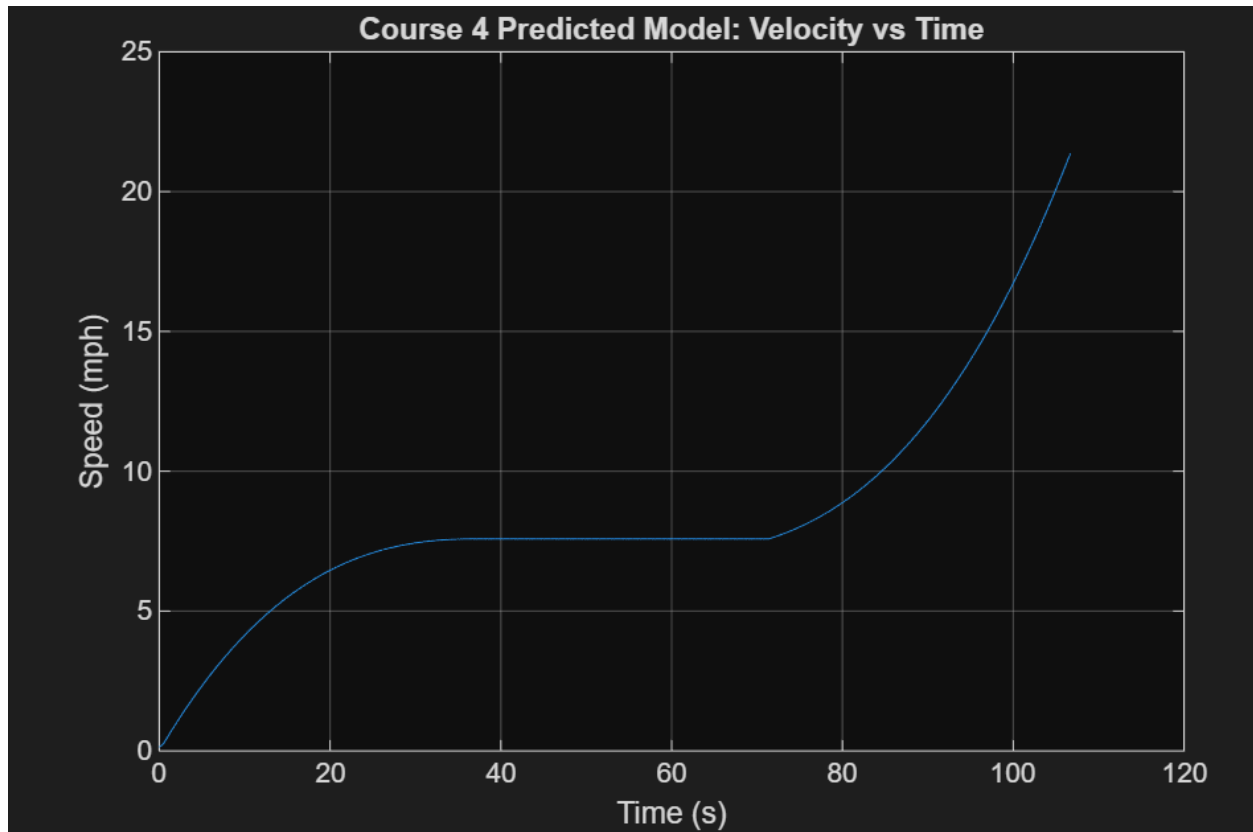


Figure 1 predicts the course 4 position over time. This was obtained by integrating the predicted velocity model with respect to time

$$x(t) = \int v(t) dt$$

The position curve is continuous, indicating that the model predicts that the vehicle is consistently moving forward without breaking.

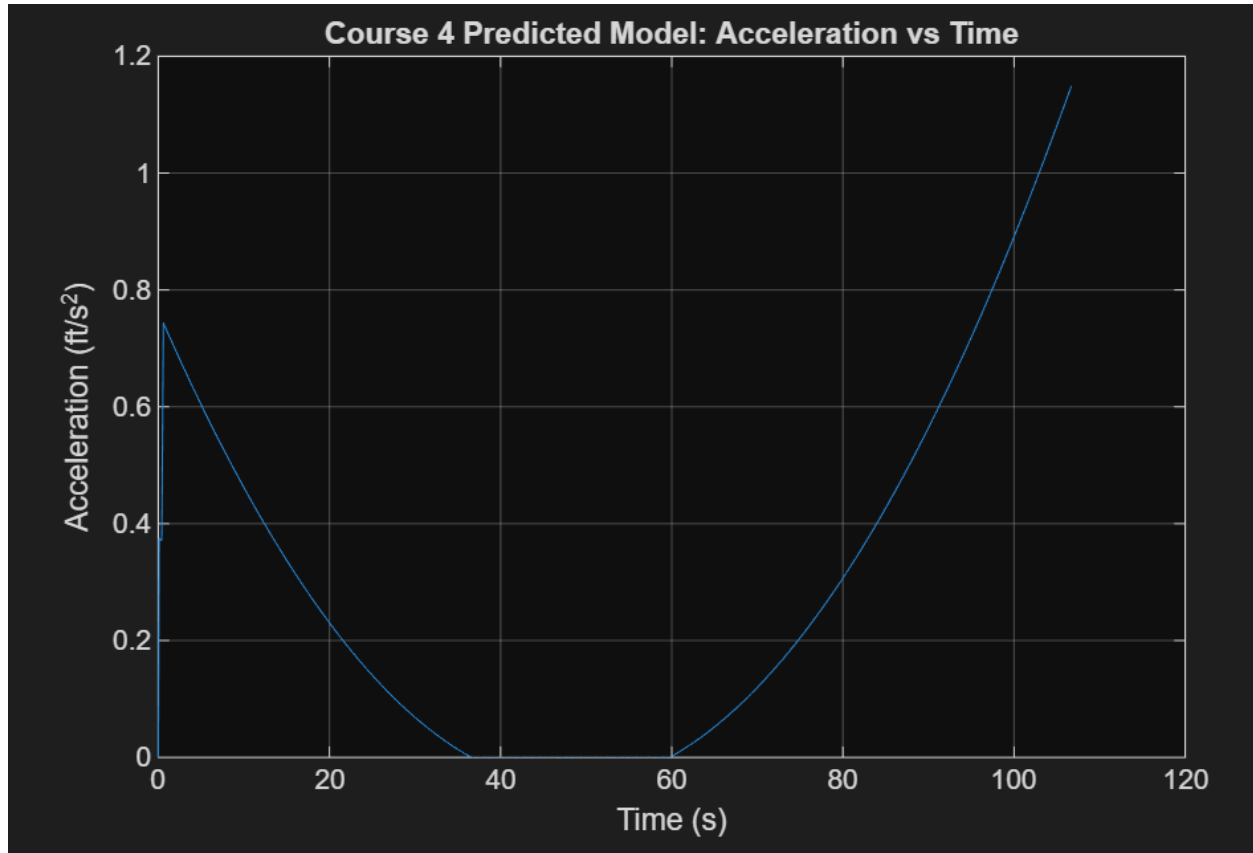
Figure 2. Velocity vs. Time



In **Figure 2**, the velocity vs. time prediction shows that the speed starts by gradually increasing, then remaining constant. Towards the end, it increases. The predicted speed remains relatively low, making the total time required to complete the quarter mile around 105 seconds. This is due to the data used to build this model. In courses 1-3, we simulated real driving data that contained low speeds. As a result, the fitted polynomial had a relatively capped acceleration behavior, leading to lower speeds and longer completion time. The cubic model was selected to represent velocity as a function of time because it captures the nonlinear acceleration behavior while remaining stable over the measured speed range.

The fit equation: $v(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0$

Figure 3. Acceleration vs. Time

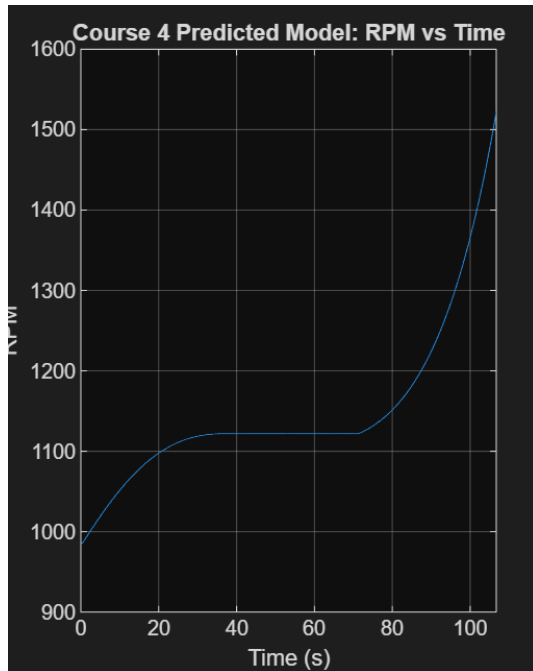


In **Figure 3**, the predicted model for acceleration vs. time was found by differentiating the velocity model with respect to time.

$$a(t) = dv/dt$$

Although acceleration data were measured directly, the acceleration model can be obtained by what we know about math and physics. To keep velocity, position, and acceleration consistent, we used derivatives and integrals to keep mathematical consistency. This accounts for uncertainties in the OBD and the different course trials. However, the measured acceleration data was still used to get acceleration phases and compare them to our model acceleration.

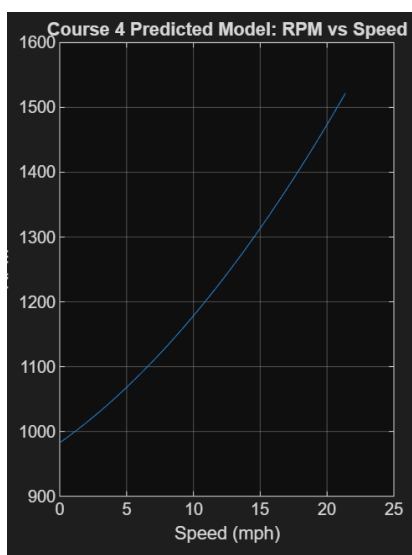
Figure 4. RPM vs. Time



$$\text{RPM} = b_2 v^2 + b_1 v + b_0$$

In **Figure 4**, it displays a predicted model of the RPM for Course 4. This is a result of the predicted velocity graph. RPM starts at 1000, showing the RPM while the car is idling, and increases as the vehicle accelerates. It remains constant when speed is constant, increases as speed increases.

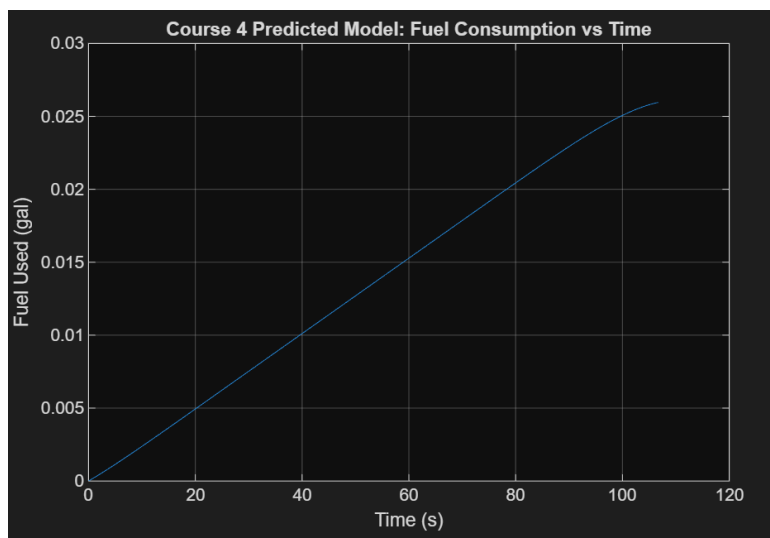
Figure 5. RPM vs. Speed



$$\text{RPM} = b_2 v^2 + b_1 v + b_0$$

Figure 5 model was created by fitting a quadratic relationship between RPM and speed. As speed increases, RPM increases since a higher engine rotational speed is required. This model predicts real-world driving behavior, rather than high-performance and high-speed models. The quadratic relationship allows us to model RPM vs. speed to capture the smooth increase in engine RPM, avoiding oscillations that can result with higher order models.

Figure 6. Fuel Consumption vs. Time



$$\text{FuelRate} = c_2 v^2 + c_1 v + c_0$$

Figure 6 indicates that the fuel rate is relatively similar. Because the vehicle travels at relatively low speeds, the fuel rate remains fairly constant. This leads to a linear fuel consumption.

Table 3. Predicted Data for Course 4

Course	Time (s)	Max Speed (mph)	Max Acceleration (ft/s ²)	Fuel Used (gal)
4	106.7	21.375	1.1493	0.025956

Vehicle speed, engine RPM, and fuel rate used low-order polynomial fits. The third-order polynomial was used for the velocity-time model. This was chosen to show the nonlinear acceleration and to avoid overfitting. Second-order polynomials were used for RPM vs. speed, fuel rate vs. speed relationships because it resulted in smooth trends. Higher-order polynomials were not used because they caused unrealistic oscillations. Using second-order and third-order keeps the model simple and stable.

Safe Driving Conditions

https://drive.google.com/file/u/1/d/1Z-s1CbAXHC2HjsLFzUIeqb5IQp49NoN_/view?usp=sharing

[g](#)

<https://drive.google.com/file/d/1q5EWQ7B7ypQKzBrCtETDSPeyKF7QdScM/view?usp=sharing>

[g](#)

https://youtube.com/shorts/mE_1QJTdgcw?si=8rd8FJfFYpYKsCg4

Discussion of Videos

Through the video documentation of the trials of each course, we demonstrated our safe, law-abiding driving practices. These videos were recorded last Tuesday, January 13th, when we were driving on the campus's Lot F closed course for motorcycle classes. Vincent's role, as

shown, is driving the vehicle while Murtaza is in the car operating the OBD reader and performing data collection. The driver is undistracted and follows the rules of the road.

Conclusion

This epic quest investigated the real-world performance of a 2018 Mazda CX-5 through experimental data collection and mathematical modeling. The primary objective was to analyze vehicle motion, engine behavior, and fuel consumption using data obtained from an OBD-II reader and smartphone-based sensor (OBD Fusion), and to compare this experimental data with models developed using MATLAB. To achieve this objective, three controlled driving courses were designed to represent common low-speed driving conditions, including acceleration, cruising, and deceleration. Course #1 involved accelerating from rest to 15 mph and maintaining that speed until reaching a distance of 100 yards. Course #2 required accelerating from rest to 25 mph and then decelerating back to 0 mph within a total distance of 200 yards. Course #3 consisted of driving a quarter mile while maintaining an average speed of 15 mph. Each course was repeated multiple times, specifically 3 rounds for each course, which improves repeatability and reduces the influence of random measurement error.

The experimental results demonstrated that vehicle motion under low-speed driving conditions can be reasonably modeled using classical kinematic relationships when supported by sufficient experimental data and appropriate numerical processing. The velocity, acceleration, and position models exhibited trends consistent with expected physical behavior, including nonlinear acceleration during initial throttle application and approximately constant deceleration during braking events. Average multiple trials improved the smoothness and reliability of the fitted models, allowing for interpolation within the tested speed range.

Engine RPM behavior showed a clear and predictable relationship with vehicle speed. The RPM versus speed models reflected the mechanical characteristics of the vehicle's transmissions and demonstrated how engine speed increases as a function of vehicle velocity under steady driving conditions. Although engine torque and power were not directly measured, the observed RPM trends aligned with expected power band behavior and provided insight into the relationship between engine performance and vehicle acceleration. Fuel consumption modeling further supported this analysis, as fuel rate increased during acceleration events and remained relatively stable during periods of constant speed. Integrating the fuel rate over time enabled estimation of total fuel usage for each driving course, illustrating how experimental data can be used to evaluate vehicle efficiency under real driving conditions.

An important component of this project was the extrapolation of the experimentally derived models to simulate vehicle performance on a quarter-mile course, representing the Irwindale Speedway scenario. Using fitted velocity, acceleration, RPM, and fuel consumption models, vehicle behavior was actually predicted beyond the original experimental range. While this extrapolation required assumptions, such as road conditions and consistent engine performance, it demonstrated the potential of data-driven models to estimate vehicle performance on a longer course. The extrapolated results produced reasonable estimates for total time, maximum speed, and fuel usage, supporting the validity of the modeling approach within the limitations of the available data.

Several sources of limitations affected the experimental results and should be considered when interpreting the findings. The use of a regular consumer-grade OBD-II reader introduced many limitations in measurement accuracy and resolution. During data collection, Bluetooth disconnections occurred between the OBD Fusion application and the OBD-II device. These

disconnections resulted in minor data interruptions and required repeated reconnections, which may have introduced timing inconsistencies and some noise into the dataset. Although repeating each course three times and averaging the results helped reduce random error, a more stable data acquisition system would have been far better for our team in this epic quest.

An additional limitation encountered during this project was the selection and configuration of a suitable OBD-II data acquisition application. Several commonly available applications were evaluated before data collection, including Torque Pro and Car Scanner. While these applications offered basic vehicle diagnostics, they were insufficient for the requirements of this epic quest due to limitations in accessible parameters, inconsistent sampling rates, or unreliable data export functionality. As a result, these applications could not consistently provide the complete set of variables needed for modeling vehicle motion, engine behavior, and fuel consumption.

To meet the epic quest requirements, the OBD Fusion application was selected, as it provided access to the necessary parameter identification (PID) data, supported a sampling rate of approximately 10 Hz, and allowed data export in CSV format for MATLAB analysis. However, this application required a one-time purchase of approximately \$10, introducing a minor financial cost associated with data collection. While this cost was relatively small, it represents an additional barrier that may not be immediately apparent when planning similar experimental studies using consumer-grade diagnostic tools. In addition to the financial consideration, the configuration and calibration of the application introduced further challenges.

The OBD Fusion application did not function optimally with default settings and required manual calibration and customization to ensure compatibility with the specific OBD-II reader used in this study. Our team was required to independently learn how to configure the application, select appropriate PIDs, adjust units, and verify that the data being recorded was

accurate and consistent across trials. This process required additional time and troubleshooting before successful data collection and introduced the possibility of configuration-related uncertainty during early test runs.

Although these challenges were ultimately resolved, they highlight an important limitation of relying on third-party consumer applications for engineering data acquisition. Differences in application design, hardware compatibility, and required calibration procedures can significantly affect data quality and repeatability. Future implementations of this project would benefit from allocating additional time for software validation, performing preliminary calibration tests, or utilizing a dedicated data acquisition system with standardized configuration and documentation.

The physical testing environment also influenced the scope of the experiment. Data collection was conducted primarily in Lot F at Mount San Antonio College, which provided a controlled and safe environment but limited the available driving distance. The compact size of the test area restricted the maximum achievable speed and reduced the duration of steady-state driving segments. Conducting future experiments in a larger area, such as Parking Lot M behind the student WIN center, would allow for longer acceleration and cruising intervals, improving the quality of the collected data and enabling more robust model validation.

Future work could expand the experimental design to include additional driving scenarios that better represent real-world operating conditions. For example, incorporating a course with significant elevation changes would allow investigation of the effects of road grade on engine RPM, acceleration, and fuel consumption. Comparing uphill acceleration with downhill coasting, without throttle input, would provide valuable insight into engine load behavior and the influence of gravitational forces on vehicle performance. Including such scenarios would strengthen the applicability of the developed models and improve their predictive capability.

Additional improvements could be made to the data processing methodology. Applying advanced filtering techniques, such as low-pass filters, could further reduce noise in acceleration and speed measurements. Integrating independent measurement sources, such as GPS-based velocity data, would allow for cross-validation of OBD-II speed readings. Increasing the sampling rate, if supported by the hardware, would also improve resolution during rapid acceleration and braking events.

The team further concluded that the predicted models provided meaningful insight into the relationships between vehicle speed, engine behavior, and fuel consumption. While individual data points exhibited variability due to sensor limitations and environmental factors, the overall model trends aligned with expected physical behavior and engineering institutions. The extrapolated quarter-mile simulation produced reasonable estimates for elapsed time, maximum speed, and fuel usage, reinforcing confidence in the modeling framework when applied within reasonable assumptions. Collectively, these findings led our team to conclude that the developed models are suitable for analyzing real-world vehicle performance at low speeds and can serve as a foundation for more advanced modeling efforts in future studies.

Overall, this epic quest successfully demonstrated how experimental vehicle data can be collected, processed, and modeled to evaluate real-world performance using engineering principles. The collaborative effort of our team allowed for the development of multiple experimental courses, repeated trials, and extrapolated simulations, resulting in a thorough analysis of vehicle dynamics under low-speed driving conditions. While limitations related to sensor accuracy, test environment size, and data connectivity were present, the results highlight how thoughtful experimental design and data analysis produce meaningful insights into vehicle performance. The methods and conclusions presented in this work provide a strong foundation

for future vehicle modeling studies and illustrate the practical application of kinematics, data analysis, and MATLAB-based modeling to a real engineering system.

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